WASP-11b

R-0725

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_241009` · created 2026-07-28T00:17:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460592.8068 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.049
Transit depth (Rp/Rs)^2	2.52 +/- 1.57 [%]
Orbital Inclination (inc)	85.76 +/- 1.55
Airmass coefficient 1 (a1)	0.6449 +/- 0.008
Airmass coefficient 2 (a2)	0.43 +/- 0.12
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	#3 - [365, 216]
Optimal Aperture	4.74
Optimal Annulus	11.14
Transit Duration (day)	0.011 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.049 vs 0.131 (deviation 0.52σ)
Duration fit vs published	0.011 d vs 0.1075 d (deviation 8.02σ)
Mid-time O–C	-7.8 min
Depth SNR	3.0
Residual scatter	1.24 %

Light curve

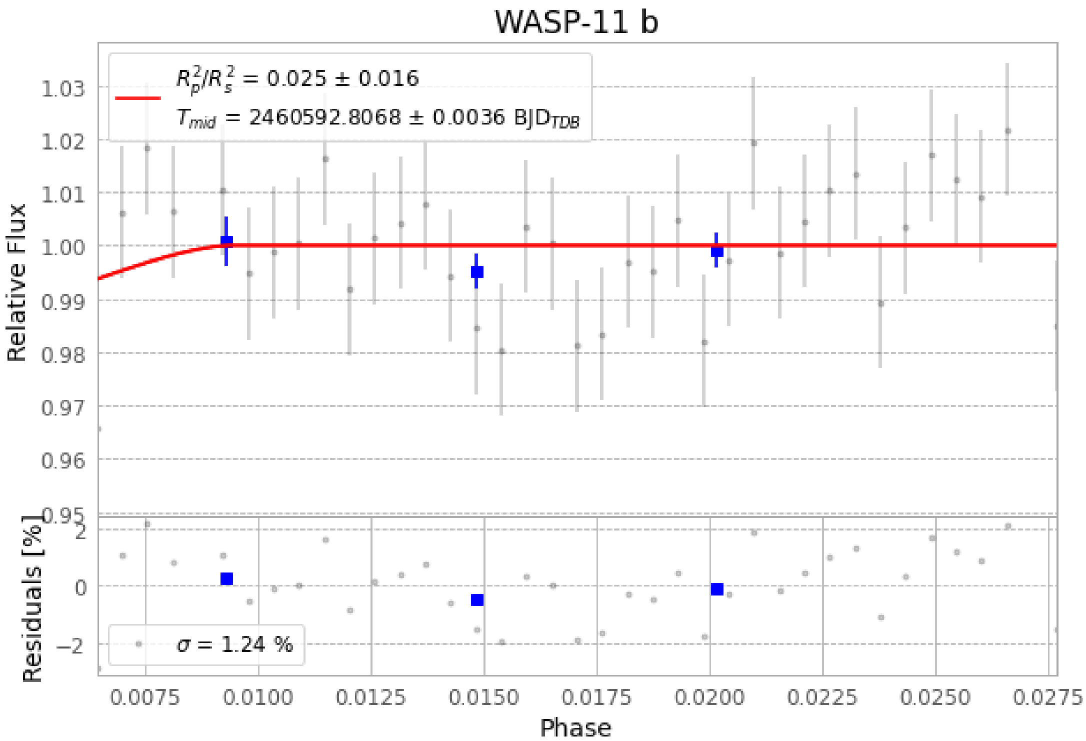

Figure 1 — FinalLightCurve_WASP-11 b_2024-10-09.png (SHA-256 b2e8443068781cf0…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [425, 344], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ae76bd31610a7e04…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

39 FITS frames (26 MB), `HATP-10241009074816.FITS` → `HATP-10241009094215.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-241009.log (7387193f2379…), FinalLightCurve_WASP-11 b_2024-10-09.png (b2e844306878…), FinalLightCurve_WASP-11 b_2024-10-09.csv (acc82b20866b…)

Compute resources

Wall time 85.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)